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WiNN enables selective correction of run-order drift and batch effects while preserving biological signal in metabolomics data

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Abstract

Motivation: Sequential mass spectrometry metabolomics data are highly susceptible to run-order drift and batch effects. Existing correction approaches depend on dense interleaved quality-control (QC) samples and apply correction broadly across metabolites without statistical evidence for feature-specific correction, increasing the risk of overcorrection and removal of genuine biological signal.
Results: We present White Noise Normalization (WiNN), a framework for selective metabolite-wise technical correction. WiNN is built on a simple inferential principle: under adequate sample randomization, correction should be applied only when residual technical structure is detectable. The method first tests each metabolite for run-order dependence within batch or inferred run segments, fits generalized additive models only for metabolites with evidence of drift, and then restricts residual batch correction to metabolites with significant batch effects, followed by conservative dilution normalization. WiNN does not require dense QC coverage and can optionally infer contiguous run segments when batch labels are unavailable. Across simulated, benchmark, and clinical serum metabolomics datasets, WiNN recovered ground-truth profiles, enabled biological marker discovery while maintaining strong class separation under low residual batch structure, and improved sample-identity structure, reference precision, and cross-plate reproducibility. These results show that WiNN achieves a strong balance between removal of technical artefacts and preservation of biological signal.
Availability: WiNN is implemented as an open-source `R` package at `https://github.com/ratschlab/winn`.
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Supplementary Information: Supplementary data are available online.

Keywords metabolomics; LC-MS; FIA-MS; batch correction; run-order drift; normalization; generalized additive models

Abbreviations FIA-MS, GAM, LC-MS, QC, WiNN

Introduction

Metabolomics using mass spectrometry-based platforms such as liquid chromatography-mass spectrometry (LC-MS) or flow injection analysis-mass spectrometry (FIA-MS) (3) provides a systems-level readout of biochemical state and is increasingly used for biomarker discovery, disease stratification, and mechanistic investigation. In a typical MS experiment, feature-level intensity profiles for samples are acquired sequentially. The resulting data matrix is high-dimensional and explicitly structured along run order, which makes it intrinsically vulnerable to acquisition-related artefacts. Ion-source contamination, column aging in LC-MS, carryover, solvent changes, instrument equilibration, and temperature fluctuations can generate gradual drift, while sample preparation and handling can introduce discrete batch shifts (1). These effects are often feature-specific and obscure biological differences that are substantially smaller than the technical variation.

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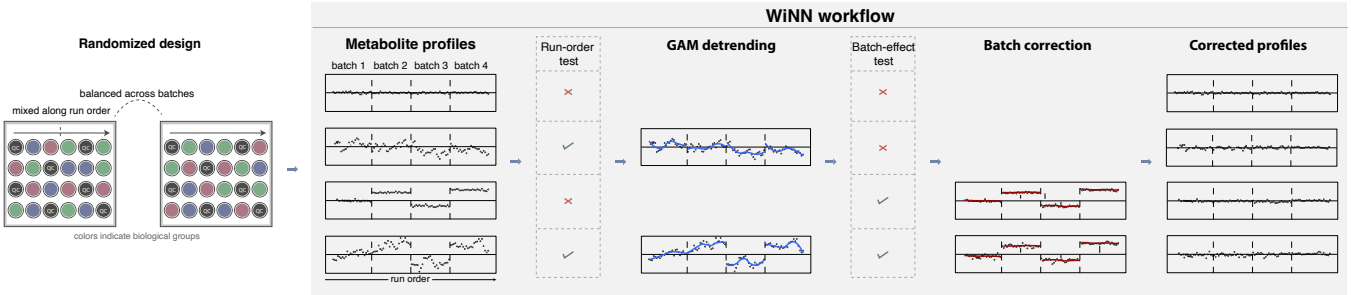

Figure 1 Overview of the WiNN workflow: randomize → test → correct. Under an appropriately randomized acquisition design, residual run-order dependence is interpreted as technical structure. WiNN tests each metabolite for run-order drift within batch or inferred run segment, applies GAM-based detrending only when drift is detected, and then applies batch correction only to metabolites with significant residual batch effects. This selective workflow reduces technical artefact while preserving biological variation.

Existing correction strategies include empirical-Bayes batch adjustment, such as ComBat, when batch labels are known, and QC-based smoothers or random-forest models that estimate run-order trends from repeated control injections (5; 1; 8). These approaches can be effective, but they are applied globally to all metabolites once a correction model has been chosen rather than after an explicit metabolite-wise assessment of whether correction is needed.

The primary safeguard remains experimental design: samples should be randomized across batches and along run order so that biological covariates are not aligned with the measurement axis (1). Without that condition, post hoc correction cannot reliably separate technical structure from biological signal. Even under appropriate randomization, normalization and batch correction involve a fundamental trade-off: insufficient correction leaves technical artefacts in place, whereas aggressive (over)correction can remove genuine biological variation, especially for metabolites with little or no detectable technical effect.

WiNN was developed to address this problem. The method implements a randomize-test-correct framework in which run-order dependence is first assessed metabolite by metabolite, drift is modeled with generalized additive models only when statistically warranted, and residual batch correction is similarly restricted to features with evidence of between-batch structure (Figure 1). WiNN is designed for sequential MS data, does not require dense QC insertion for trend fitting and can optionally infer contiguous run segments when batch labels are unavailable, providing a practical fallback when batch annotations are incomplete. This makes WiNN well suited to clinical and translational studies in which QC capacity is limited but preservation of biological heterogeneity is essential.

Materials and methods

Overview

WiNN is a selective correction framework that uses sample randomization to define a white-noise null for run-order dependence and applies correction selectively rather than uniformly, first testing for drift and then for residual batch effects (Figure 1). In its default configuration, it operates on $\log(1+x)$-transformed intensities and requires only the intensity matrix, run order, and optionally batch labels and QC identifiers.

Design assumptions and randomization

WiNN, like other post hoc correction methods, assumes that batch assignment and acquisition order are independent of the biological variables of interest. While designing experiments, samples should be distributed across batches so that biological groups or covariates are balanced and randomized within batch. This assumption is fundamental, because if phenotype is confounded with batch or run order, post hoc correction cannot reliably distinguish technical structure from true signal (1; 6).

In practice, we recommend verifying randomization during study design using batch and run-order balance diagnostics. Use χ^2 tests for batch balance, Runs tests for categorical sequence randomness, and rank von Neumann tests for ordered covariates. For complex designs with correlated or dependent samples, multivariable assessment or constrained randomization may be preferable to repeated univariate testing (6).

When feasible, pooled QC injections should be placed at regular intervals to monitor analytical stability and support evaluation or optional parameter selection; in LC-MS, one QC every 7–8 study samples is a practical recommendation (1). WiNN treats serial metabolite-specific dependence along the acquisition axis as technical structure.

Stepwise WiNN workflow

Let $X = (x_{ij}) \in \mathbb{R}_{\geq 0}^{p \times n}$ denote the metabolite intensity matrix, with metabolites $i = 1, \ldots, p$, injections $j = 1, \ldots, n$, segment labels $b(j)$, and within-segment run order t_j. WiNN implements the selective randomize–test–correct scheme shown in Figure 1, in which both drift removal and residual batch adjustment are restricted to metabolites with detectable technical structure. Detailed mathematical specifications are provided in the Supplementary Methods S1-5.

1. **Preprocessing.** Metabolite-wise isolated extremes are regularized by robust median/MAD shrinkage and the data are analyzed on the $\log(1 + x)$ scale. Shrinkage is used in place of outlier exclusion to limit undue leverage on downstream inference while preserving informative extreme observations.
2. **Segment-wise drift testing.** Within each annotated or inferred segment, WiNN tests each metabolite for serial dependence along run order under a design-based white-noise null. By default, this uses the Ljung–Box test

with segment-adaptive lag selection; a Durbin–Watson alternative is available for first-order dependence. Within each segment, metabolite-wise p-values are controlled by the Benjamini–Hochberg procedure.

3. **Selective detrending.** Only metabolites with significant run-order dependence are detrended. For these, WiNN fits a generalized additive model (GAM) in run order using spline-based smoothers implemented in `mgcv` (11) and subtracts the centered fitted effect,

$$z_{ij}^{\mathrm{drift}} = z_{ij} - \left[\hat{f}_i^{(s)}(t_j) - \bar{\hat{f}}_i^{(s)} \right],$$

where z_{ij} is the transformed intensity, $\hat{f}_i^{(s)}$ is the segment-specific smooth for metabolite i, and $\bar{\hat{f}}_i^{(s)}$ is its segment mean. This removes ordered structure while preserving the segment location. Standard mode uses cubic regression splines; conservative mode uses a more strongly penalized low-rank smoother with simplified fallbacks for short or unstable segments.

4. **Selective residual batch correction.** After detrending, WiNN tests each metabolite for residual between-batch effects using a one-way batch model with FDR control and applies batch correction only to significant metabolites. A ComBat-based empirical-Bayes alternative may also be used on the transformed scale.
5. **Conservative dilution normalization.** Corrected values are back-transformed and normalized by probabilistic quotient normalization (PQN) using the cohort median spectrum, or the control median when pooled controls are available, as reference. By default, WiNN applies clipped shrinkage of sample-wise dilution factors, thereby reducing residual dilution effects without aggressive global normalization.

Automatic parameter detection

When pooled controls are available, WiNN can select analysis settings by grid search over the enabled drift-testing, smoothing, batch-correction, and dilution-normalization options. Candidate pipelines are scored using control-only agreement and variability criteria, so controls are used for model selection and evaluation, not for direct drift fitting. We use this mode throughout the paper unless stated otherwise, and refer to it simply as WiNN.

Automatic batch detection

When batch labels are unavailable, WiNN can optionally infer contiguous run segments before correction using functional pruning optimal partitioning (fkPELT), applied to signal along run-order under a penalized likelihood objective (9). The resulting changepoints define inferred segments which are used for downstream drift testing, detrending, and residual batch assessment; when segments are inferred from the same data, the subsequent p-values are interpreted conditionally on that estimated segmentation and are not intended as a full post-selection error-control guarantee.

Comparison methods

We benchmarked WiNN against five representative correction methods spanning the main classes of adjustment used in sequential MS metabolomics: ComBat for empirical-Bayes batch correction (5), QC-RLSC for QC-guided local-regression drift correction (1), QC-RFSC for QC-trained random-forest correction (8), SERRF for multivariate QC-based random-forest normalization (2), and TIGER for ensemble-learning-based technical correction (4). For the competing non-WiNN methods, parameters were tuned by grid search to maximize benchmark performance; implementation details, package versions, and the ranking procedure are provided in the Supplementary Methods.

Evaluation datasets

Simulation benchmark

WiNN was first evaluated on synthetic LC-MS-like data with known ground truth. Each dataset comprised 1,000 metabolites across 500 injections arranged as five plates of 100 injections, with pooled control injections interleaved regularly within plate. Ground-truth abundances were generated on the log scale and transformed to positive intensity space. Technical variation was then added as metabolite-specific plate offsets and independent within-plate drift processes. Batch effects included both unaffected metabolites and a subset with stronger plate-to-plate shifts. Drift effects were generated separately for each metabolite within each plate from a mixture of linear, nonlinear, autoregressive, and composite trajectories with varying amplitudes. This design induces both discrete plate effects and realistic within-plate run-order structure. Full generative details are given in the Supplementary Methods S6.

Public multi-batch measurement benchmark

We next analyzed a public MetaboLights benchmark for repeated multi-batch measurement (7). The dataset contains cow and sheep heart samples together with controls, providing clear biological separation across sample classes. The same study samples were measured across eight analytical batches, yielding eight observations per sample under distinct batch conditions. Injection order was recorded and pooled QC injections were interleaved throughout acquisition. This design tests whether correction reduces technical structure while preserving class separation and improving consistency across repeated measurements.

Internal randomized FIA-MS ToF serum cohort

Finally, we evaluated WiNN in an internal clinical serum metabolomics cohort acquired by FIA-MS ToF using a 96-well plate design optimized to separate technical from biological variation. Study samples were distributed across plates under explicit randomization constraints, while preserving within-patient longitudinal structure by placing time-course samples adjacently but in an order unrelated to collection time. To reduce confounding between acquisition order and sample metadata, plate layouts were iteratively re-randomized and assessed using χ^2 tests across plates and plate blocks and Runs tests on within-plate sample sequences. Pooled QC material, generated from a random subset of study samples, was inserted every eighth well, and each biological sample was prepared on two plates and injected twice per plate, yielding four technical measurements per sample. Blank wells and NIST1950 reference injections were also included on each plate to assess analytical stability, repeatability, and residual plate effects. This design provides a stringent benchmark for correction of plate-associated and run-order artefacts while preserving between-sample heterogeneity.

Table 1. Simulation benchmark. Held-out QC CV was computed on QC samples excluded from correction. Residual run-order structure is summarized as mean GAM deviance explained across non-QC metabolite-segment profiles. Ground-truth agreement was computed on non-QC samples. Values are mean ± SD, rounded to two decimals; lower is better for QC CV and Residual run-order deviance.

Method	Held-out QC CV	Residual run-order deviance	Metabolite ICC(A,1)	Sample Pearson
Raw	42.37 ± 26.10	0.12 ± 0.19	0.66 ± 0.20	0.58 ± 0.02
ComBat	12.99 ± 10.04	0.12 ± 0.19	0.92 ± 0.08	0.90 ± 0.03
QC-RLSC	19.87 ± 26.34	0.03 ± 0.05	0.88 ± 0.11	0.74 ± 0.03
QC-RFSC	11.83 ± 9.29	0.10 ± 0.12	0.93 ± 0.06	0.69 ± 0.05
TIGER	10.17 ± 10.55	0.08 ± 0.11	0.94 ± 0.06	0.89 ± 0.02
SERRF	12.58 ± 9.44	0.09 ± 0.14	0.92 ± 0.08	0.89 ± 0.02
WiNN	**8.59 ± 6.06**	**0.02 ± 0.03**	**0.95 ± 0.03**	**0.94 ± 0.01**

Evaluation metrics

Across datasets, we used complementary metrics to quantify technical artefact removal and preservation of biological structure. On the simulation benchmark, evaluation used held-out QC coefficient of variation (CV), residual run-order structure quantified as mean GAM deviance explained by run order, metabolite-level agreement with ground truth, and sample-level agreement with ground truth. Ground-truth agreement was summarized using metabolite-profile ICC(A,1) and sample-profile Pearson correlation. For the real-data benchmarks, biological and technical structure were summarized by the proportion of variance explained by sample type, sample identity, batch, or plate. On the public benchmark, biological recovery was additionally quantified by the number of sample-type-associated metabolites, together with held-out QC CV, feature-level agreement, sample-replicate agreement, and residual run-order deviance. Importantly, although held-out QC labels were used for evaluation, `WiNN` (auto) and other methods did not have access to these held-out QC labels during model selection or correction. On the internal clinical cohort, repeated measurements and held-out NIST1950 reference injections enabled evaluation of sample-replicate agreement, cross-plate feature agreement, independent reference precision, and residual run-order structure. Complete metric definitions are provided in the Supplementary Methods S8.

Results

Ground truth recovery in simulated data

We first used simulated data with known ground truth to verify that WiNN removed injected technical structure while preserving the underlying metabolite and sample profiles. In this setting, WiNN had the best values on all four primary metrics (Table 1).

WiNN improved ground-truth agreement at both the metabolite and sample levels, increasing metabolite-profile ICC(A,1) from 0.66 ± 0.20 in the raw data to 0.95 ± 0.03 and sample-profile Pearson correlation from 0.58 ± 0.02 to 0.94 ± 0.01. It also achieved the lowest held-out QC coefficient of variation (CV), reducing this from 42.37 ± 26.10 in the raw data to 8.59 ± 6.06, ahead of the next-best method, TIGER, at 10.17 ± 10.55. Residual run-order structure, quantified by mean GAM deviance explained, fell from 0.12 ± 0.19 in the raw data to 0.02 ± 0.03 after WiNN correction.

Post-correction Ljung–Box testing was consistent with these results. Of 5,000 metabolite-plate profiles, WiNN left significant residual autocorrelation in only 30, whereas the second-best method, QC-RFSC, left 637; TIGER, SERRF, and QC-RLSC left 814, 927, and 2,205, respectively, while ComBat left this quantity unchanged because it does not model within-batch drift. This indicates that WiNN closely recovered the ground truth and removed the injected drift more completely than the alternative methods.

These results indicate that WiNN recovered ground-truth structure, improved held-out technical precision, and reduced residual drift relative to the comparison methods. Additional WiNN operating modes are shown in the Supplementary Results S1.

Biological signal preservation in a public multi-batch benchmark

We next evaluated WiNN on the public MTBLS79 benchmark (7), testing whether technical correction could reduce batch structure without suppressing biological signal. In the raw data, batch dominated the leading principal components (Figure 2A), explaining 36.1% of the variance, whereas sample group explained 34.3%. After WiNN correction, batch-associated variance fell to 1.1% and sample-group variance increased to 46.7% (Figure 2B–D), the best values for these metrics among the compared methods. Without supplied batch labels, automatic batch inference reduced batch-associated variance to 2.36% while sample-group variance remained 46.37% (Figure 2C,D), although performance was weaker than with supplied labels.

This pattern was consistent with downstream marker recovery. WiNN identified 1,003 sample-group-associated metabolites, the highest number among the comparison methods and 13 more than the raw data (Figure 2E).

QC-RFSC achieved the minimum held-out QC CV, but WiNN and ComBat performed equally well on feature-level ICC(A,1) and sample-replicate Pearson at the reported precision, while WiNN had the lowest Residual run-order deviance among all methods (Table 2). Taken together, these metrics indicate that some methods reduced QC variability more aggressively, whereas WiNN left less residual technical structure while preserving more biological signal. Notably, the default no-QC mode remained close to auto(QC) in simulation and public data: in simulation the two modes were numerically indistinguishable across the primary metrics, and in the public benchmark no-QC gave similar technical metrics with slightly higher sample-group

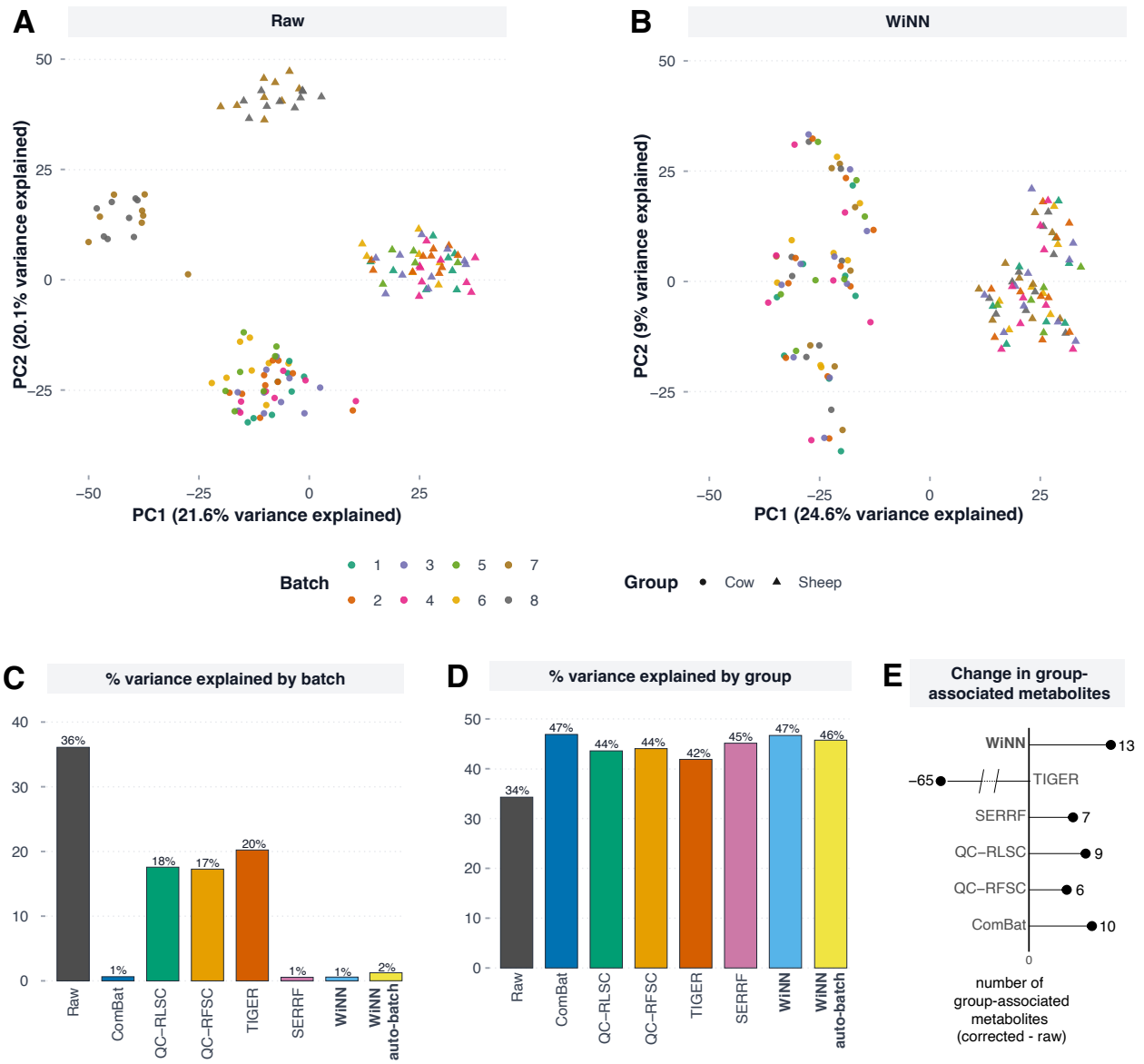

Figure 2 Biological signal preservation in the public multi-batch benchmark. (A) PCA of the raw data, showing strong batch-associated structure. (B) PCA after WiNN correction, showing removal of batch structure and clearer biological grouping. (C) Proportion of variance explained by batch across methods, including WiNN with automatic batch inference. (D) Proportion of variance explained by sample group across methods. (E) Change in the number of sample-group-associated metabolites relative to the raw data. Together, these results indicate reduced technical structure with preserved biological signal recovery after WiNN correction.

variance explained and metabolite detection (Supplementary Tables S1–S3).

Sample-identity preservation in a clinical FIA-MS cohort

The internal clinical FIA-MS cohort provided the most stringent benchmark, as each biological sample was measured repeatedly across plates and independent NIST1950 reference injections were available for orthogonal performance assessment. Relative to the alternative methods, `WiNN` was better at both maximizing sample-identity variance explained and minimizing plate-associated variance, with and without batch information, increasing sample-identity variance explained from 65.8% in the raw data to 77.8% while reducing plate-associated variance from 13.9% to 0.11%.(Figure 3A,B). Relative to the strongest alternative methods, WiNN retained more sample-specific structure than SERRF and slightly more than TIGER, while leaving substantially less residual plate structure than either. Downstream biological analyses of this clinical cohort are beyond the scope of the present study.

This pattern was also supported by independent technical validation. `WiNN` achieved the lowest held-out NIST1950 CV, a marginally higher cross-plate metabolite ICC(A,1) than competing methods, tied for the best sample-replicate Pearson correlation, and slightly lower residual run-order deviance than all other methods (Table 3). The clinical benchmark therefore showed simultaneous gains in reference precision, cross-plate

Table 2. Public dataset benchmark. Held-out QC CV was computed on QC injections excluded from QC-based training. Residual run-order structure is summarized as mean GAM deviance explained across metabolite–segment profiles. Values are mean ± SD.

Method	Held-out QC CV	Residual run-order deviance	Cross-batch Metabolite ICC(A,1)	Sample-replicate Pearson
Raw	29.13 ± 19.43	0.14 ± 0.16	0.51 ± 0.31	0.95 ± 0.01
ComBat	16.10 ± 9.48	0.14 ± 0.16	**0.60 ± 0.31**	**0.98 ± 0.00**
QC-RLSC	17.15 ± 9.62	0.15 ± 0.17	0.54 ± 0.32	0.96 ± 0.01
QC-RFSC	**7.53 ± 4.14**	0.15 ± 0.17	0.54 ± 0.32	0.85 ± 0.03
TIGER	14.29 ± 9.13	0.14 ± 0.16	0.53 ± 0.33	0.96 ± 0.01
SERRF	18.57 ± 12.32	0.14 ± 0.16	0.58 ± 0.30	0.97 ± 0.00
WiNN	15.96 ± 9.85	**0.13 ± 0.16**	**0.60 ± 0.31**	**0.98 ± 0.00**

Table 3. Clinical FIA-MS cohort. Held-out NIST1950 CV measures precision on independent reference injections. Residual run-order structure is summarized as mean GAM deviance explained across metabolite–segment profiles. Values are mean ± SD.

Method	Held-out NIST1950 CV	Residual run-order deviance	Cross-plate metabolite ICC(A,1)	Sample-replicate Pearson
Raw	17.91 ± 10.99	0.12 ± 0.14	0.53 ± 0.24	0.98 ± 0.02
ComBat	16.16 ± 7.79	0.12 ± 0.14	0.58 ± 0.22	0.98 ± 0.01
QC-RLSC	24.07 ± 27.36	0.13 ± 0.13	0.52 ± 0.23	0.80 ± 0.08
QC-RFSC	18.88 ± 11.07	0.11 ± 0.11	0.56 ± 0.24	0.76 ± 0.15
TIGER	15.89 ± 10.65	0.08 ± 0.09	0.54 ± 0.24	0.99 ± 0.01
SERRF	15.25 ± 10.16	0.10 ± 0.12	0.58 ± 0.23	0.99 ± 0.01
WiNN	**13.03 ± 4.79**	**0.07 ± 0.10**	**0.59 ± 0.22**	**0.99 ± 0.01**

agreement, and residual run-order structure without loss of sample-replicate agreement.

Discussion

Across simulation, public data, and the internal clinical cohort, WiNN reduced technical structure while retaining biological or sample-specific structure across the evaluated benchmarks. WiNN was designed to preserve biological or sample-specific structure while removing ordered and batch-associated artefact, and the benchmarks support that objective across ground-truth, multi-batch, and repeated-measure settings.

This distinction is important because correction in metabolomics is inherently an identifiability problem, not simply a smoothing problem. Much of the field still treats drift and batch adjustment as a downstream transformation applied once a correction model has been selected, often without first establishing that a given feature exhibits statistically detectable technical structure. That strategy can work when artefacts are strong and homogeneous, but it is harder to justify when drift is weak, absent, or feature-specific, and it increases the risk of attenuating genuine biological variation. WiNN addresses this by making statistical evidence a prerequisite for correction and by restricting both drift modeling and residual batch adjustment to metabolites that exceed prespecified significance criteria. The aim is therefore not minimal residual variability per se, but a better-controlled trade-off between artefact removal and signal preservation.

The results also reinforce the importance of experimental design. WiNN, like other post hoc correction methods, relies on the assumption that batch assignment and acquisition order are not aligned with the biological variables of interest. Under valid randomization, residual run-order dependence can be interpreted as technical structure. When this assumption is violated, residual run-order structure may reflect biology rather than technical variation, and selective correction may attenuate true signal. In addition, the current automatic batch-detection step is designed for contiguous acquisition-scale shifts and may be less effective for more complex or partially overlapping technical structure; in our benchmarks this mode was generally weaker than using supplied labels. Finally, performance differences relative to strong baselines were benchmark-dependent.

A practical advantage of WiNN is lower computational overhead than newer QC-based machine-learning methods. In our benchmarks, WiNN was consistently faster than SERRF, TIGER, and QC-RFSC while remaining competitive on the main signal-preservation endpoints (Supplementary Table S6). This is relevant for larger studies, where repeated model fitting across many metabolites can become limiting.

Several extensions are worth pursuing. A cluster-aware variant that borrows information across correlated metabolites could improve power in short segments while retaining the selective logic of the current framework. More broadly, the present results suggest that the approach should extend naturally beyond LC-MS to sequential MS settings more generally, including FIA-MS and targeted assays, where ordered drift and discrete batch effects coexist. In that sense, the broader implication is methodological: normalization should be framed less as how aggressively one can correct a dataset and more as when correction is statistically justified for a given feature.

Supplementary Data

Supplementary data are available at *Bioinformatics* online.

Data availability

WiNN is available as an open-source `R` package at `https://github.com/ratschlab/winn`. Instructions to generate the simulated datasets, download the public MetaboLights dataset, and reproduce the analyses and figures are available at `https://github.com/ratschlab/WiNN-for-technical-correction`. The internal clinical dataset is not publicly available because of patient privacy and consent restrictions.

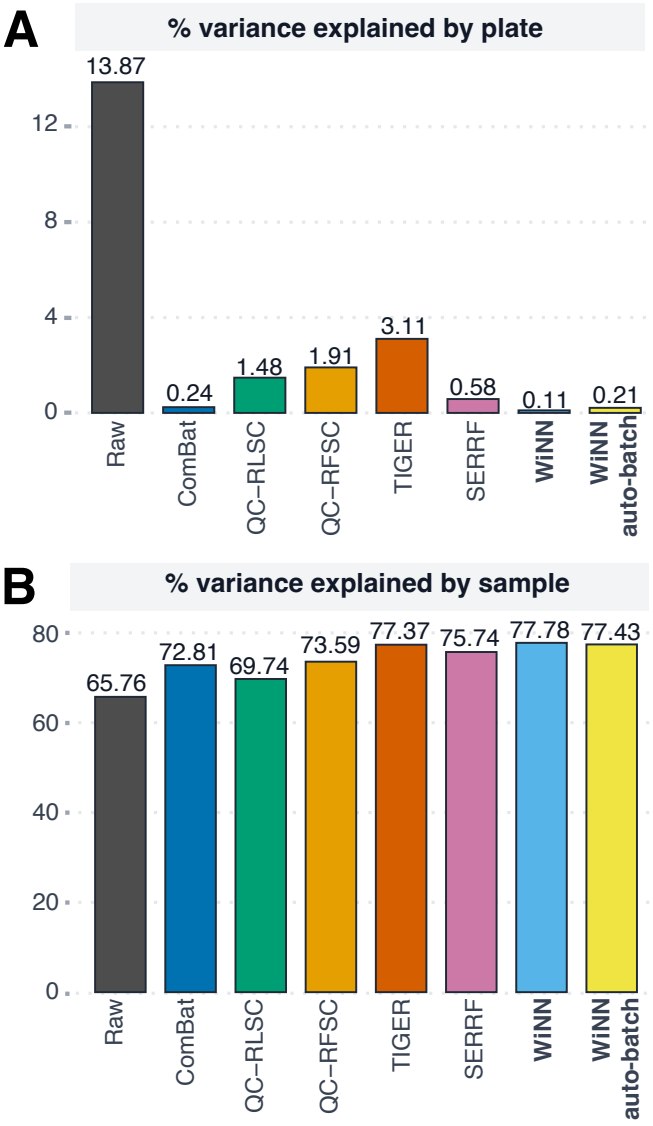

Figure 3 Sample-identity preservation in the clinical FIA-MS cohort. (A) Proportion explained by plate; lower values indicate less residual plate structure. (B) Proportion of variance explained by sample identity; higher values indicate stronger retained sample structure. WiNN combined high sample-identity structure with low plate-associated variance.

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Author contributions

T.T. (Conceptualization [lead], Formal analysis [lead], Methodology [lead], Software [lead], Writing—original draft [lead], Funding acquisition [supporting]), Y.Z. (Formal analysis [supporting], Methodology [supporting], Software [supporting]), A.S. (Methodology [supporting], Software [supporting]), F.G. (Conceptualization [supporting]), M.L. (Conceptualization [supporting], Methodology [supporting]), A.J. (Methodology [supporting], Software [supporting]), H.L.G. (Conceptualization [supporting]), A.K. (Funding acquisition [equal]), M.P.L (Funding acquisition [supporting]), G.R. (Funding acquisition [lead], Supervision [equal]), and O.D. (Conceptualization [equal], Methodology [equal], Supervision [lead]).

Conflicts of interest

The authors declare no conflicts of interest.

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Supplementary Methods and Results

Supplementary Methods

S1. Workflow, white-noise rationale, and outlier regularization

The full WiNN workflow is shown in Supplementary Figure S1. Under an appropriately randomized acquisition design, residual structure aligned with run order is interpreted as technical rather than biological, so WiNN tests for metabolite-specific deviations from a white-noise null and applies correction selectively rather than globally.

Let $X = (x_{ij}) \in \mathbb{R}_{\geq 0}^{p \times n}$ denote the metabolite intensity matrix, with metabolites $i = 1, \ldots, p$, injections $j = 1, \ldots, n$, run order t_j, and batch or segment label $b(j)$.

WiNN first regularizes isolated extremes metabolite-wise using the median and median absolute deviation (MAD). For metabolite i, let

$$m_i = \mathrm{median}_j(x_{ij}), \qquad d_i = \mathrm{MAD}_j(x_{ij}),$$

and define thresholds

$$L_i = m_i - 4d_i, \qquad U_i = m_i + 4d_i.$$

Values within $[L_i, U_i]$ are unchanged. Upper-tail outliers are shrunk by linear interpolation from the upper non-outlier range toward $[m_i + 3d_i, U_i]$. Lower-tail outliers are treated analogously, mapping extreme lower values toward $[L_i, m_i - 3d_i]$. The aim is to reduce leverage of isolated extremes without deleting observations. Downstream testing is performed on the transformed scale

$$z_{ij} = \log(1 + \tilde{x}_{ij}),$$

where $\tilde{x}_{ij}$ denotes the outlier-adjusted intensity.

S2. Drift testing and selective GAM detrending

Within each annotated or inferred segment s of length n_s, WiNN tests each metabolite for serial dependence along run order. The default test is the Ljung–Box statistic

$$Q_i^{(s)}(h_s) = n_s(n_s + 2) \sum_{\ell=1}^{h_s} \frac{\hat{\rho}_{i\ell}^{(s)\,2}}{n_s - \ell},$$

where $\hat{\rho}_{i\ell}^{(s)}$ is the empirical lag-ℓ autocorrelation of the log-transformed profile. The associated p-value is evaluated against a χ^2 distribution with $h_s - d$ degrees of freedom, where the implementation uses $d = 1$ for the Ljung–Box branch.

If the lag is not user-specified, WiNN chooses it adaptively as

$$h_s = \min\left\{n_s - 1,\ \max\left(\min\left(10, \left\lfloor \frac{n_s}{5} \right\rfloor\right),\ d + 3\right)\right\}.$$

Thus, the default Ljung–Box lag is segment-adaptive, bounded above by 10, and skipped when the segment is too short to support the required degrees of freedom.

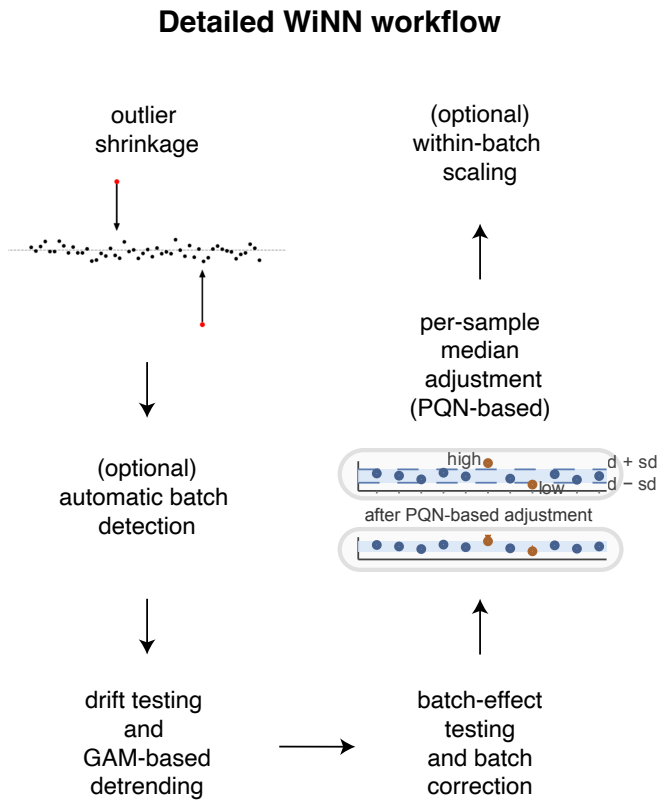

Figure S1 Detailed overview of the steps in the WiNN workflow

As an alternative, WiNN can use a Durbin–Watson test for first-order serial dependence. For metabolite i in segment s, this corresponds to testing residual autocorrelation after fitting

$$z_{ij} = \beta_{0i}^{(s)} + \beta_{1i}^{(s)} t_j + e_{ij}^{(s)},$$

with Durbin–Watson statistic

$$D_i^{(s)} = \frac{\sum_{j=2}^{n_s} \left(e_{ij}^{(s)} - e_{i,j-1}^{(s)}\right)^2}{\sum_{j=1}^{n_s} \left(e_{ij}^{(s)}\right)^2}.$$

Within each segment, metabolite-wise p-values are adjusted by Benjamini–Hochberg. Thus, the controlled family is the set of metabolite-level drift hypotheses within that segment, not the union of such hypotheses across all segments.

In the default selective mode, only metabolites with adjusted drift p-value below the chosen FDR threshold are detrended. For such metabolites, WiNN fits a smooth function of run order and subtracts the centered fitted effect,

$$z_{ij}^{\mathrm{drift}} = z_{ij} - \left\{\hat{f}_i^{(s)}(t_j) - \overline{\hat{f}_i^{(s)}}\right\}, \qquad \overline{\hat{f}_i^{(s)}} = \frac{1}{n_s} \sum_{j \in s} \hat{f}_i^{(s)}(t_j),$$

thereby removing ordered structure while preserving the segment mean on the log scale. Corrected values are then returned to

intensity space as

$$x_{ij}^{\text{drift}} = \max\left\{\exp\left(z_{ij}^{\text{drift}}\right) - 1,\, 0\right\}.$$

Standard detrending uses `mgcv::gam` with a cubic regression spline,

$$z \sim s(t, \text{bs} = \texttt{"cr"}).$$

The conservative implementation uses a more defensive fitting sequence. For segments with $n_s < 5$, WiNN falls back directly to linear regression. Otherwise it sets

$$k_s = \max\left(4, \min\left(10, \left\lfloor \frac{n_s}{5} \right\rfloor\right)\right)$$

and attempts, in order: penalized P-splines with REML and $\gamma = 1.4$, thin-plate splines with REML and $\gamma = 1.2$, fixed-degree cubic regression splines with $k = \min(6, k_s)$, then LOESS, and finally a linear model if smoother fitting fails.

If `detrend_non_autocorrelated = "spline"`, the white-noise gate is bypassed and all metabolites are detrended within each segment.

S3. Residual batch correction, PQN adjustment, and optional scaling

After drift correction, WiNN tests each metabolite for residual between-batch structure on the log scale using the one-way model

$$z_{ij}^{\text{drift}} = \mu_i + \beta_{i,b(j)} + \varepsilon_{ij}.$$

This yields a metabolite-wise ANOVA F-test, with Benjamini–Hochberg adjustment across metabolites. Only significant metabolites are corrected in the ANOVA branch. For those metabolites, WiNN applies mean-preserving residualization,

$$z_{ij}^{\text{batch}} = z_{ij}^{\text{drift}} - \bar{z}_{i,b(j)}^{\text{drift}} + \bar{z}_i^{\text{drift}},$$

where $\bar{z}_{i,b(j)}^{\text{drift}}$ is the batch-specific mean and $\bar{z}_i^{\text{drift}}$ is the global mean for metabolite i. This removes batch-specific location shifts while preserving the overall feature mean.

As an alternative, WiNN can apply ComBat on the $\log(1+x)$ scale using an intercept-only design matrix. This approach uses the empirical-Bayes ComBat model with default parametric prior and returns corrected values after inverse transformation and non-negativity clipping.

WiNN then performs optional PQN-based dilution adjustment in intensity space. Let x_{ij}^{batch} denote the post-batch-correction matrix and define the reference spectrum

$$r_i = \begin{cases} \text{median}_{j \in C}\, x_{ij}^{\text{batch}}, & \text{if pooled controls } C \text{ are provided,} \\ \text{median}_j\, x_{ij}^{\text{batch}}, & \text{otherwise.} \end{cases}$$

Sample-wise quotients and dilution factors are

$$q_{ij} = \frac{x_{ij}^{\text{batch}}}{r_i}, \qquad d_j = \text{median}_i(q_{ij}).$$

In full normalization mode,

$$x_{ij}^{\text{norm}} = \frac{x_{ij}^{\text{batch}}}{d_j}.$$

In the default shrinkage mode, only extreme dilution factors are adjusted. Let $\bar{d}$ and s_d be the mean and standard deviation of $\{d_j\}$. Then

$$x_{ij}^{\text{norm}} = \begin{cases} x_{ij}^{\text{batch}} \dfrac{\bar{d} - s_d}{d_j}, & d_j < \bar{d} - s_d, \\ x_{ij}^{\text{batch}} \dfrac{\bar{d} + s_d}{d_j}, & d_j > \bar{d} + s_d, \\ x_{ij}^{\text{batch}}, & \text{otherwise.} \end{cases}$$

This conservative PQN step leaves central samples unchanged and shrinks only unusually low or high dilution factors toward the bulk distribution. If `median_adjustment = "none"`, this step is skipped.

Finally, WiNN can optionally standardize each metabolite within each batch:

$$x_{ij}^{\text{scaled}} = \frac{x_{ij}^{\text{norm}} - \mu_{i,b(j)}}{\sigma_{i,b(j)}},$$

where $\mu_{i,b(j)}$ and $\sigma_{i,b(j)}$ are the within-batch mean and standard deviation. Zero standard deviations are replaced by 1 in the implementation.

S4. QC-based automatic parameter detection

In the default fixed-parameter mode, WiNN uses MAD-based outlier regularization, selective drift correction based on the Ljung–Box test with adaptive lag selection and FDR threshold 0.05, conservative spline detrending for metabolites flagged as non-white-noise along run order, ANOVA-based mean-only batch correction at FDR threshold 0.05, PQN-based shrinkage normalization, and no within-batch scaling. When pooled controls are provided and `parameters = "auto"`, WiNN selects analysis settings by grid search using control-only agreement and variability metrics. The exact grid depends on the enabled options. If batch labels are supplied, the batch source is fixed to the provided labels; otherwise WiNN first auto-detects segments and uses those inferred labels downstream. The grid then spans the supplied autocorrelation test options, spline method (`conservative` or `standard`), drift FDR threshold $\{0.10, 0.05, 0.01\}$ in selective mode, batch-correction FDR threshold $\{0.10, 0.05, 0.01\}$ in the ANOVA branch, and PQN mode (`shrink` or `normalize`). The within-batch scaling flag is not optimized over both states; it follows the user-specified setting.

For each candidate pipeline, WiNN computes the mean pairwise correlation among controls,

$$\bar{r}_C = \frac{2}{|C|(|C|-1)} \sum_{j<k,\; j,k \in C} \text{cor}(x_{\cdot j}, x_{\cdot k}),$$

and the mean feature-wise control coefficient of variation,

$$\overline{\text{CV}}_C = \frac{1}{p} \sum_{i=1}^{p} \frac{\text{sd}(x_{iC})}{|\,\text{mean}(x_{iC})|}.$$

The score is

$$S = \bar{r}_C - 0.5\, \overline{\text{CV}}_C,$$

and the parameter set with maximal S is selected. Controls are therefore used for parameter selection and evaluation, but not for fitting run-order trends themselves.

For efficiency, WiNN caches structural computations: drift corrections are prepared once per batch/test/smoother setting and then materialized across multiple FDR thresholds, and the same strategy is used for ANOVA-based batch correction.

S5. Automatic batch detection by fkPELT

When batch labels are unavailable, WiNN infers contiguous run segments before downstream correction. It first computes a robust sample-level summary

$$a_j = \mathrm{median}_i(\tilde{x}_{ij}),$$

that is, the sample-wise median signal after outlier adjustment.

Changepoints are then estimated by minimizing a penalized segmentation objective,

$$\sum_{k=0}^{K} \mathcal{C}\big(a_{(\tau_k+1):\tau_{k+1}}\big) + \beta K,$$

over

$$0 = \tau_0 < \tau_1 < \cdots < \tau_K < \tau_{K+1} = n.$$

The implementation uses an fkPELT dynamic-programming scheme with pruning. For candidate segments of length at least five, $\mathcal{C}$ is a Gaussian negative log-likelihood based on a natural-spline mean fit in run index; for shorter segments, the cost reduces to a constant-mean Gaussian model. Internal spline knots are placed sparsely, at approximately one per 60 injections, to favor broad acquisition-scale changes.

If the penalty is numeric, that value is used directly. Otherwise WiNN resolves either

$$\beta_{\mathrm{BIC}} = \hat{\sigma}^2 \log(n)$$

or the more conservative default

$$\beta_{\mathrm{MBIC}} = 3\log(n),$$

where $\hat{\sigma}^2$ is estimated robustly from first differences of $\{a_j\}$ using a MAD-based variance estimator. The resulting changepoints define the contiguous inferred segments used as batch labels in all downstream drift and batch steps. When these segments are inferred rather than supplied, the subsequent drift and batch tests are conditional on the estimated segmentation and should be interpreted accordingly rather than as fully post-selection-adjusted inferential guarantees.

S6. Simulation design

Synthetic benchmarking data were generated to provide both a clean reference matrix and a matched artefact-bearing matrix for downstream method evaluation. The simulations comprised $p = 1{,}000$ metabolites measured across $n = 500$ injections arranged on five plates of 100 injections each. Pooled control injections were inserted every 10 acquisitions within each plate, yielding 10 controls per plate and 50 controls overall. Separate random seeds were used for ground-truth generation, batch-effect assignment, batch-effect magnitude, drift assignment, drift-type allocation, and drift-shape generation so that individual stochastic components could be modified without changing the entire simulation. No correction method was applied during data generation itself; the goal was to create transparent benchmark inputs with known underlying structure.

Ground-truth signal was generated on the log scale as

$$G_{ij}^{(\log)} \sim \mathcal{N}(0,\, 0.35^2),$$

independently across metabolites and injections, and then mapped to positive intensity space by

$$G_{ij} = \exp\big(G_{ij}^{(\log)}\big).$$

To impose a known technical-replicate structure, all pooled control injections were set equal to a single anchor profile on the log scale before technical effects were added. Thus, in the ground truth, controls were exact technical replicates and any loss of control agreement in the observed matrix arose solely from the simulated technical artefacts.

Batch effects were added on the log scale to a subset of metabolites. Specifically, 70% of metabolites were designated batch-affected. For each affected metabolite, plate-specific offsets were sampled independently across the five plates and then centered to have zero mean across plates, so that the perturbation represented a relative plate effect rather than a global abundance shift. Most affected metabolites received moderate offsets sampled from

$$\mathcal{N}(0,\, 0.6^2),$$

whereas 10% of batch-affected metabolites were assigned a stronger regime with offsets sampled from

$$\mathcal{N}(0,\, 1.35^2).$$

These centered plate offsets were then applied to all injections belonging to the corresponding plate.

Drift was introduced independently at the metabolite-plate level. First, 60% of metabolites were designated drift-susceptible. For each such metabolite, each plate was marked drift-active with probability 0.55, with at least one active plate enforced for every drift-susceptible metabolite. Each active metabolite–plate pair was then assigned one of four drift morphologies: linear (35%), nonlinear (30%), autoregressive AR(1) (15%), or mixed (20%). Drift trajectories were generated as functions of normalized within-plate run position, centered to zero mean, and rescaled so that their maximum absolute deviation equaled an amplitude drawn uniformly from 0.20 to 0.85 on the log scale. Linear drift used a random slope; nonlinear drift combined a sinusoidal component with a localized Gaussian bump; AR(1) drift was generated from an autoregressive process with coefficient sampled uniformly from 0.45 to 0.90; and mixed drift combined linear, nonlinear, and autoregressive components.

The final observed log-intensity matrix was constructed as

$$Y_{ij}^{(\log)} = G_{ij}^{(\log)} + B_{i,b(j)} + D_{ij},$$

where $B_{i,b(j)}$ denotes the metabolite- and plate-specific batch offset and D_{ij} denotes the simulated drift term. The observed intensity matrix was then obtained by exponentiation,

$$Y_{ij} = \exp\big(Y_{ij}^{(\log)}\big).$$

The simulation therefore produced paired clean and technically perturbed datasets, together with sample metadata and a

feature-level design table recording whether each metabolite was assigned batch effects, stronger plate shifts, and drift on one or more plates.

S7. Public multi-batch dataset preprocessing and imputation

For the MetaboLights dataset, missing-value handling was performed before correction and downstream benchmarking. Metabolites observed in fewer than 80% of samples were removed and missing entries were imputed using `missForest` (10) with a fixed random seed (`set.seed(42)`), and the imputed feature matrix was used for subsequent analyses.

S8. Evaluation metric definitions

Performance was assessed using complementary metrics for technical artefact removal and preservation of biological structure.

Held-out QC precision.
Precision on pooled QC or reference injections was summarized by the metabolite-wise coefficient of variation,

$$\mathrm{CV}_i = \frac{\mathrm{sd}(x_{iC})}{|\,\mathrm{mean}(x_{iC})|},$$

where x_{iC} denotes intensities for metabolite i across the relevant held-out control set C. Reported QC precision was the mean CV across metabolites. The same definition was used for held-out NIST1950 reference injections.

Residual run-order structure.
Residual technical structure along acquisition order was quantified at the *metabolite-segment* level. Specifically, within each batch/plate/contiguous run segment, a generalized additive model of intensity versus run order was fitted for each metabolite, and the deviance explained by run order was recorded as

$$\mathrm{DevExp}_{i,s} = 1 - \frac{D^{\mathrm{res}}_{i,s}}{D^{\mathrm{null}}_{i,s}},$$

where $D^{\mathrm{res}}_{i,s}$ is the residual deviance of the fitted model for metabolite i in segment s, and $D^{\mathrm{null}}_{i,s}$ is the deviance of the corresponding intercept-only model. Reported residual run-order structure was summarized across all metabolite-segment fits for a given method, primarily as the mean deviance explained (with corresponding dispersion summaries); lower values indicate weaker remaining run-order dependence.

Metabolite-level agreement.
Metabolite-level agreement was assessed by representing metabolite i as its abundance profile across injections,

$$x_{i\cdot} = (x_{i1}, \ldots, x_{in}).$$

Agreement was then quantified using the two-way mixed-effects absolute-agreement intraclass correlation coefficient, ICC(A,1), on matched metabolite profiles. On the simulation benchmark, this compared the corrected and ground-truth profiles,

$$\mathrm{ICC}^{\mathrm{met}}_i = \mathrm{ICC}(\mathrm{A},1)\big(\hat{x}_{i\cdot},\, x^*_{i\cdot}\big)\,,$$

with summary performance given by

$$\overline{\mathrm{ICC}}^{\mathrm{met}} = \frac{1}{p}\sum_{i=1}^{p} \mathrm{ICC}^{\mathrm{met}}_i.$$

On real-data benchmarks without ground truth, the same ICC(A,1) framework was applied to matched metabolite profiles across repeated, paired, or cross-plate measurements, with results again summarized as the mean across metabolites.

Sample-level agreement.
Sample-level agreement was assessed by representing injection j as its abundance profile across metabolites,

$$x_{\cdot j} = (x_{1j}, \ldots, x_{pj}).$$

Agreement was quantified by Pearson correlation on matched sample profiles. On the simulation benchmark, this compared each corrected sample to its ground-truth counterpart,

$$r^{\mathrm{samp}}_j = \mathrm{cor}\big(\hat{x}_{\cdot j},\, x^*_{\cdot j}\big)\,,$$

with summary performance

$$\bar{r}^{\mathrm{samp}} = \frac{1}{n}\sum_{j=1}^{n} r^{\mathrm{samp}}_j.$$

On real-data benchmarks with repeated measurements of the same biological sample, sample-level reproducibility was quantified analogously as the Pearson correlation between matched replicate sample profiles across metabolites and summarized across replicate pairs.

Biological signal recovery.
On the public benchmark, biological recovery was additionally summarized by the number of metabolites associated with sample type under the downstream differential analysis procedure. Specifically, replicate measurements for each metabolite were first averaged across plates within each sample, after which sample types were compared using Welch's two-sample t-test on these sample-level means. Resulting p-values were adjusted across metabolites using the Benjamini–Hochberg procedure. Metabolites with FDR-adjusted $p < 0.05$ were counted as sample-type-associated, such that larger counts indicate stronger recovery of expected biological structure.

Variance explained by biological and technical factors.
For real-data benchmarks, biological and technical structure were summarized using a variance-weighted principal-component R^2 score. Briefly, principal component analysis was performed on the $\log(1+x)$-transformed data matrix, and for each of the first five principal components PC_k, the proportion of variance explained by the variable of interest z (e.g. sample type, sample identity, batch, or plate) was quantified by the coefficient of

determination

$$R_k^2 = R^2(PC_k \sim z).$$

These component-specific values were then aggregated using the variance explained by each component as weights,

$$R^2_{\text{weighted}} = \sum_{k=1}^{K} w_k R_k^2, \qquad w_k = \frac{\lambda_k}{\sum_{\ell=1}^{K} \lambda_\ell},$$

where λ_k is the variance explained by PC_k and $K = \min(5, \text{number of available PCs})$. Higher weighted R^2 for biological variables indicates stronger biological signal retention, whereas lower weighted R^2 for technical variables indicates more effective artefact removal.

Competing method implementations and tuning.
The competing methods were run using ComBat from `sva` version 3.58.0, QC-RLSC from `qcrlscR` version 0.1.3, QC-RFSC from `statTarget` version 1.40.0, TIGER from `TIGERr` version 1.0.2, and SERRF from `malbacR` version 0.1.0. For the non-WiNN methods, tuning was performed by parameter grid search on a reduced feature subset; candidate configurations were then ranked separately on the evaluation metrics used for the benchmark, these ranks were summed to form a composite score, and the configuration with the best total rank was used for full-data evaluation.

Supplementary Results

S1. Benchmark-wise comparison of WiNN operating modes

Table S1. Simulation benchmark comparison across three WiNN operating modes. Values are mean ± SD. Lower is better for held-out QC CV and residual run-order deviance; higher is better for cross-batch metabolite ICC(A,1) and sample-replicate Pearson.

Method	Held-out QC CV	Residual run-order deviance	Metabolite ICC(A,1)	Sample Pearson
WiNN auto (QC)	8.59 ± 6.06	0.018 ± 0.031	0.934 ± 0.053	0.940 ± 0.007
WiNN auto-batch (QC)	12.11 ± 6.75	0.088 ± 0.129	0.886 ± 0.081	0.898 ± 0.070
WiNN default (no QC)	8.60 ± 6.05	0.018 ± 0.031	0.934 ± 0.053	0.940 ± 0.007

Table S2. Public multi-batch benchmark comparison across three WiNN operating modes. Lower is better for held-out QC CV and residual run-order deviance; higher is better for cross-batch metabolite ICC(A,1) and sample-replicate Pearson.

Method	Held-out QC CV	Residual run-order deviance	Cross-batch Metabolite ICC(A,1)	Sample-replicate Pearson
WiNN auto (QC)	15.99 ± 9.84	0.0.1338 ± 0.1582	0.60 ± 0.31	0.98 ±0.00
WiNN auto-batch (QC)	17.32 ± 11.23	0.0.1380 ± 0.1632	0.59 ± 0.31	0.97 ± 0.00
WiNN default (no QC)	16.68 ± 9.84	0.1341 ± 0.1587	0.59 ± 0.31	0.97 ± 0.00

Table S3. Public multi-batch benchmark comparison across three WiNN operating modes for removal of batch effects and preservation of biological signal.

Method	Batch variance explained (%)	Sample/group variance explained (%)	Number of metabolites detected
WiNN auto (QC)	1.12	46.68	1003
WiNN auto-batch (QC)	2.36	46.37	986
WiNN default (no QC)	1.09	47.15	1013

Table S4. Internal clinical FIA-MS cohort comparison across three WiNN operating modes. Lower is better for held-out QC CV and residual run-order deviance; higher is better for cross-plate metabolite ICC(A,1) and sample-replicate Pearson.

Method	Held-out QC CV (NIST1950)	Residual run-order deviance	Cross-plate Metabolite ICC(A,1)	Sample-replicate Pearson
WiNN auto (QC)	13.03 ± 4.79	0.07 ± 0.10	0.59 ± 0.22	0.99 ± 0.01
WiNN auto-batch (QC)	13.08± 4.93	0.04 ± 0.06	0.57 ± 0.21	0.99 ± 0.01
WiNN default (no QC)	14.81 ± 5.68	0.01 ± 0.02	0.56 ± 0.21	0.99 ± 0.01

Table S5. Internal clinical FIA-MS cohort comparison across three WiNN operating modes for removal of batch effects and preservation of biological signal.

Method	Plate variance explained (%)	Sample-identity variance explained (%)
WiNN auto (QC)	0.11	77.78
WiNN auto-batch (QC)	0.21	77.43
WiNN default (no QC)	0.10	73.63

S2. Runtime comparison.

Table S6. End-to-end runtime comparison across datasets. Reported values are total runtimes in minutes and therefore include parameter tuning, when performed, together with subsequent correction. Simulation runtimes were converted from seconds to minutes for consistency across datasets.

Method	Simulation benchmark	Public multi-batch benchmark	Internal clinical FIA-MS cohort
Raw	0.00	0.00	0.00
ComBat	0.02	0.04	0.18
QC-RLSC	0.05	0.06	0.36
QC-RFSC	0.58	1.31	4.45
TIGER	11.80	47.02	25.97
SERRF	2.14	14.38	15.67
WiNN auto (QC)	0.38	0.54	2.93
WiNN auto-batch (QC)	0.43	0.54	4.31
WiNN default (no QC)	0.05	0.04	0.34